MyLearna
Plan • Capture • Grow

MATHEMATICS
GEOMETRY AND SPATIAL REASONING

Name: ____________________

Date: ____________________

Use Geometry to Model and Interpret Space

Step 11

Upper Elementary

Today I am learning to use geometry to model and interpret space.

1. Design a Classroom Layout

You are designing a classroom. Include:

- Teacher desk
- Student tables
- Reading corner
- Storage area

a) Draw the layout on the grid.

b) Label each area.

c) Explain why you placed each item where you did.

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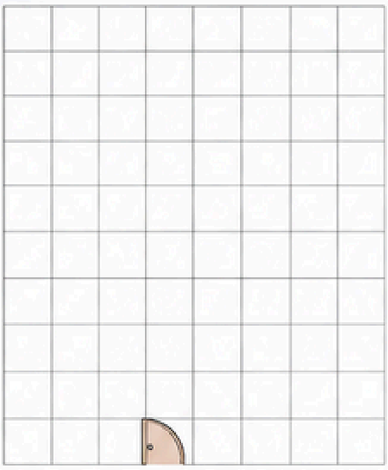
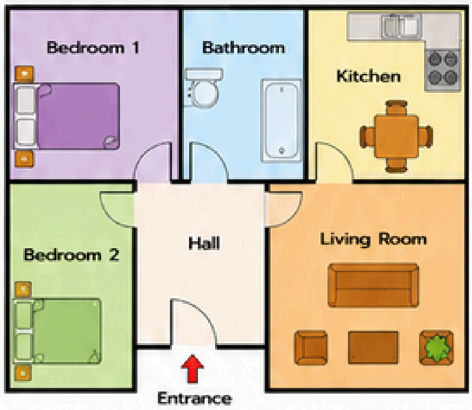


2. Floor Plan Investigation

Look at the floor plan.



a) Which room has the largest space?

____

b) Which room is closest to the entrance?

____

c) Which room would be easiest to access?

____

d) Suggest one improvement to the floor plan.

____

____

3. Geometry in Architecture

Look at the building.

a) Rectangles
Number = ________

b) Triangles
Number = ________

c) Squares
Number = ________

d) Why might architects use these shapes?

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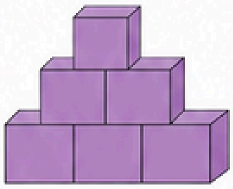
4. Build and Interpret a Structure

Look at the block structure.



a) Draw the front view.

b) Draw the side view.

c) Draw the top view.

5. Scale Drawing Challenge

Scale: 1 square = 1 metre

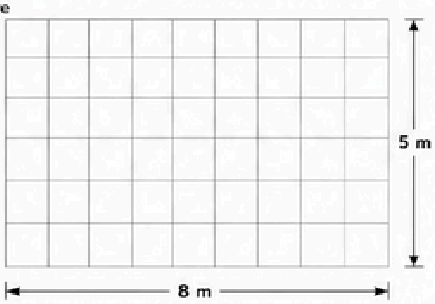
A playground is:

- 8 m long
- 5 m wide

a) Draw the playground on the grid.

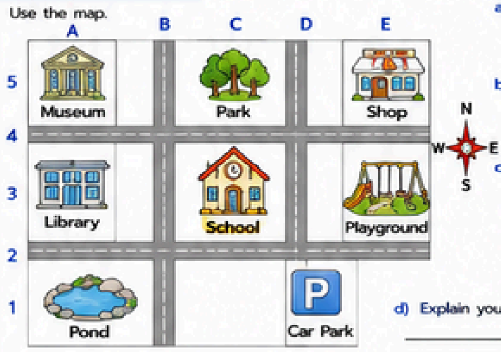
b) Add:
- slide
- swing
- tree

c) Label measurements.



6. Map Interpretation

Use the map.



a) What is north of the school?

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b) What is west of the library?

____

c) Which route is shortest from the school to the park? Show the route on the map.

____

d) Explain your reasoning.

____

____

____

7. Spatial Design Challenge

Design a small park.

Requirements:

- ☐ Pond
- ☐ Playground
- ☐ Walking path
- ☐ Trees

a) Create a layout.

b) Use labels.

c) Explain your design choices.

____

____

8. Think and Explain

a) How does geometry help us design spaces?

____

____

b) Why are floor plans useful?

____

____

c) How can geometry help solve real-world problems?

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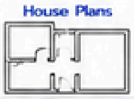
I can...

- ☐ design and model spaces.
- ☐ read and interpret maps and layouts.
- ☐ use scale and geometry in real situations.
- ☐ explain my thinking.

Geometry in Real Life!

City Plans

House Plans



Maps

Landscapes

Great job!
You are a geometry problem solver!